# Lecture XII - Metro Passenger Flow Control

Applied Optimization with Julia

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## Introduction

### FIFA World Cup 2022 in Qatar

#### Learning Objectives

After this lecture, you will be able to:

- **Model** time-expanded passenger flows with precomputed delays
- **Translate** a safety goal into inflow control constraints
- **Explain** how the set $\mathcal{R}_{e,t}$ connects periods and minutes
- **Embed** an optimization model into a rolling-horizon heuristic

#### Public transport

- FIFA World Cup 2022 took place in a **small region**
- The capacity of the metro system was **finite**
- More than **1 million tourists** were expected
- Metro usage was **free for all ticket holders**
- Transport methods were expected to be **overloaded**

**Question: What could become an issue?**

### Crowd Disasters

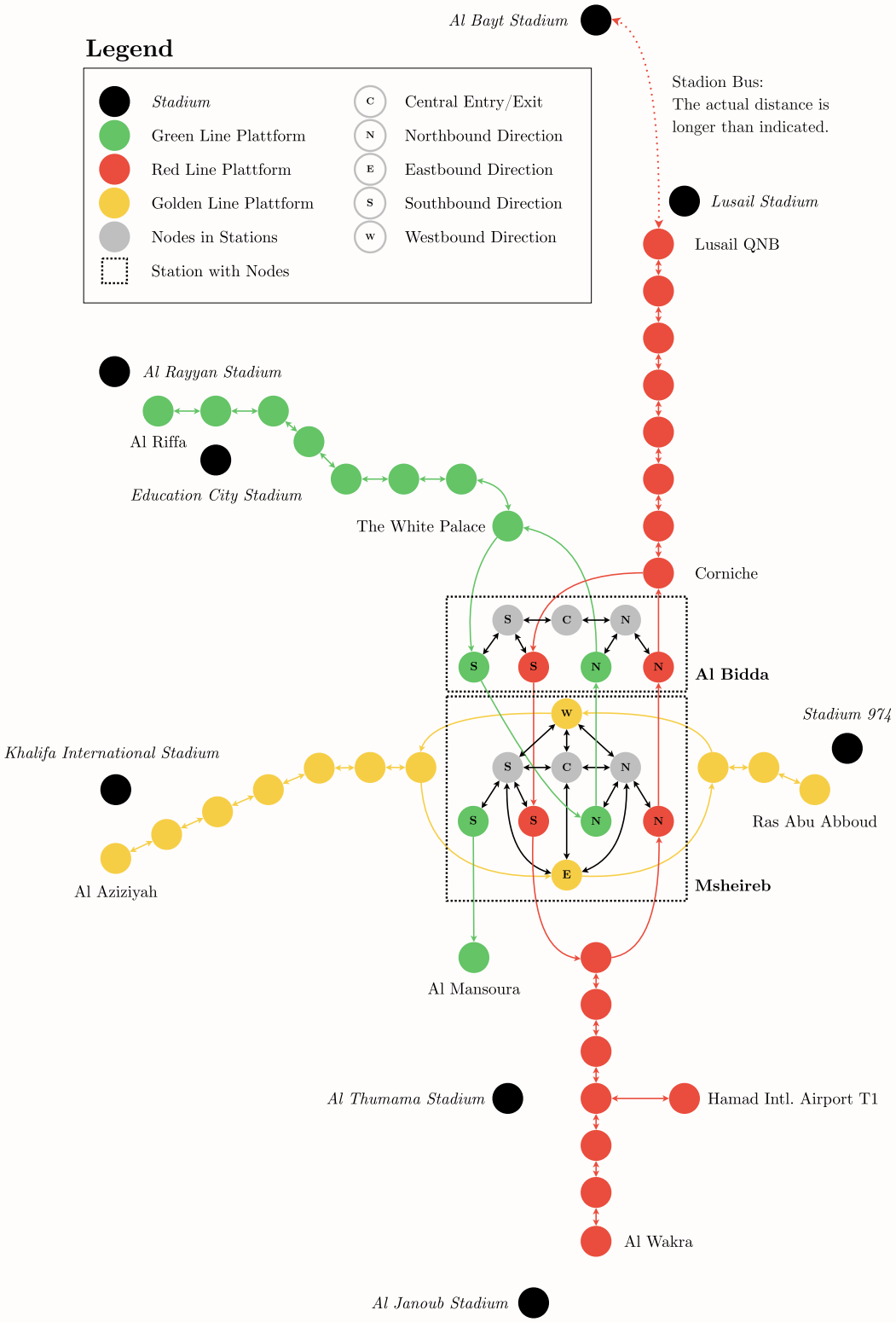
#### Potential locations

**Question: Where could crowd disasters happen?**

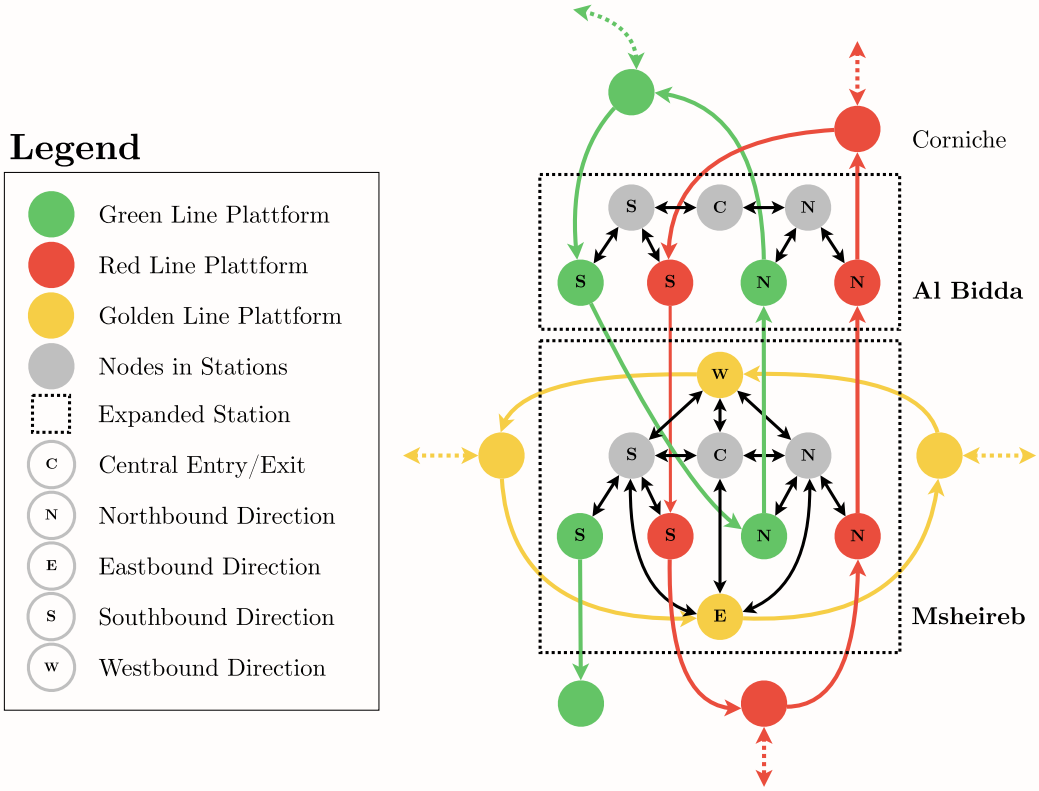
- At event venues and in waiting areas
- At intersections of multiple pedestrian flows
- In narrow passages and entrances
- On crowded metro platforms and transfer stations
- At ticket/turnstile bottlenecks and emergency exits

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# Metrosystem of Doha



## Closer look at the Metro System




## Main Bottleneck

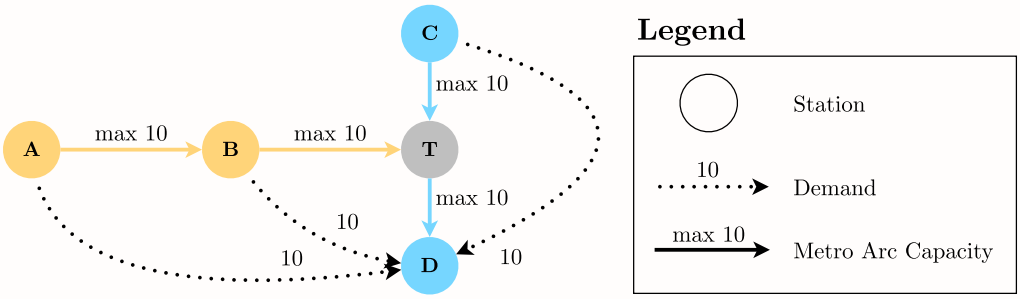
**Question: What could be a potential bottleneck?**

- Red Line has twice the capacity of Green and Gold Line
- People use the metro to get to the event venues

**Question: What could be the issue?**

- Imagine a match at a stadium along the Gold line
- People from Red and Green line will try to get there
- This will cause a **massive crowd at transfer stations**

## Visualization




- 10 people want to start per minute at **each station**
- Everybody wants to get to **station D**

## Capacities inside the stations

- Capacities **inside the stations** are limited as well
- These include the escalators, stairs, and elevators
- But also the platforms and the ticket gates

> ⚠ **Warning**
> This can lead to **overcrowding and potential crowd disasters!**

## General Issues

**Question: What could also be an issue?**

- Often unknown **how many people** exactly will participate
- Gathering data is **extremely important and difficult**
- Crowd behavior can be **unpredictable and dynamic**
- Weather conditions may affect **transportation preferences**
- Cultural factors influence **crowd movement patterns**
- Emergency situations require **flexible contingency plans**

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Let's take a

closer look at the

problem structure!

# Problem Structure

## Objective

**Question: What could be the objectives of the authorities?**

- A **safe and successful event** as host country
- Good **publicity and a positive recognition** worldwide
- **Satisfied visitors** that enjoyed their time

> **Note: We can model none of the above directly!**
> But we can assist in the estimation of transport demands and the design of operational plans to ensure public **safety** and a **smooth transportation** through the city.

## Underlying Problem

**Question: Can we just start modeling?**

- First, we need to understand the **movement patterns**
- Event is **unprecedented**, movement patterns are unknown
- Multiple **concurrent events** affect flow patterns

> **Tip: We need to estimate the data first!**
> This is a **huge challenge**, but we can use **simulation** to estimate the data.

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## Simulation

- Based on **publicly available data** from the area
- Simulates **all individuals** participating at the event
- Includes **all transportation infrastructure**
- **Individual mode choice** based on a choice model

> **Tip: Julia is a great tool for this!**
> The simulation was built in Julia. With 1,000,000 individuals walking, using cars, buses, and the metro, simulating a day took **less than 5 minutes**.

## Results of the Simulation



- Detailed movement patterns throughout the region
- Potential sections at risk in the transport infrastructure
- Potential capacity overloads at event locations

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But it is still

based on a lot

of assumptions!

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# Main risk: Metro

## Idea

**Question: What can we do to prevent crowd disasters?**

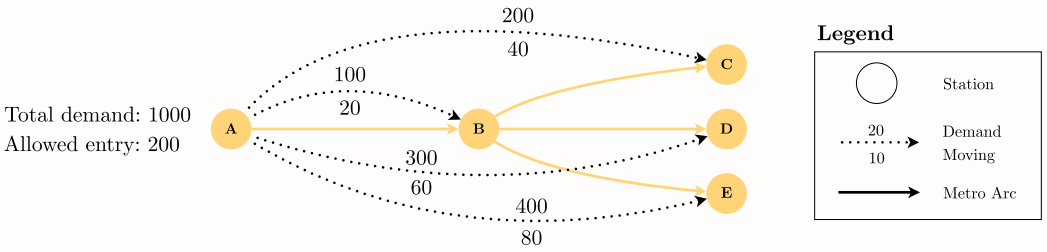
- Regulate the **inflow** at each individual station

- Ensure utilized capacity is **always within bounds**

**Question: What could we try to model?**

- **Minimize the queues** outside of the metro stations
- Based on the **allowed inflow** and origin-destination data
- Adhering to the **capacity constraints**

## Difficulty




- Patterns have different origin-destination pairs
- Regulating the inflow does not affect the destination

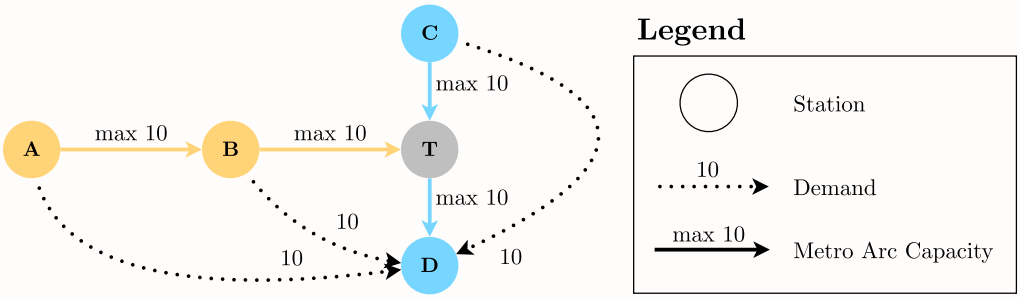
## Your Turn

Play: https://beyondsimulations.github.io/Applied-Optimization/games/metro.html

# Model Formulation

## Graph Sets?

**Question: What could be the sets for the graph?**




## Graph Sets?

- $\mathcal{G}$ - Connected digraph of the metro network $\mathcal{G} = (\mathcal{O}, \mathcal{E})$
- $\mathcal{O}$ - Set of metro stations, indexed by $o$

- $\mathcal{E}$ - Set of directed arcs between connected stations

> **Note: These are the easier sets.**
> These simply help us to represent the entire metro system as a graph with different nodes $\mathcal{O}$ and arcs $\mathcal{E}$ between connected stations.

## Time Sets?

**Question: What could be the time sets?**

- $\mathcal{T}$ - Set of minutes in the time horizon, indexed by $t$
- $\mathcal{P}$ - Set of periods in the observed time horizon, where $p \in \{1, 2, ..., n\}$

> **Note: Number of periods**
> We further define $n$ as the **number of periods in observed time horizon.**

## Periods

**Question: Why do we add periods here?**

- Staff needs **clear**, consistent instructions
- Frequent changes in flow **increase risk of errors**
- **Easier** to manage and communicate for planners

> **Note: Period Length**
> We define the period length as the length of the period in minutes measured as $m$ minutes, which is the **same for all periods**.

## Mapping Minutes to Periods

- We can define an **additional set** with their relation
- It specifies the relation of **periods $p$ to minutes $t$.**

$$I_p = \{t \in \mathcal{T} | (p-1) \times m + 1 \leq t \leq p \times m\} \quad \forall p \in \mathcal{P}$$

**Question: Can anybody explain it?**

- $I_p$ is the set of minutes $t$ that belong to period $p$
- Minutes are **not overlapping**, but they are **continuous**

## Parameters?

**Question: What could be possible parameters?**

- $q_{o,d,p}$ - Queued passengers from station $o$ to $d$ with $o, d \in \mathcal{O}$ in $p$
- $\tau_e$ - Travel time (min) of the arcs $e \in \mathcal{E}$
- $c_e$ - Max. allowed arc entry rate $e$ per minute with $e \in \mathcal{E}$
- $c_o^{min}$ - Min. station entry rate $o$ per minute with $o \in \mathcal{O}$
- $c_o^{max}$ - Max. station entry rate $o$ per minute with $o \in \mathcal{O}$
- $\alpha$ - Safety factor applied to all capacities $(0 < \alpha < 1)$

## Metro Movement

**Question: How do people move inside the metro?**

- Simple network: assume people use the **shortest path**
- Use **Dijkstra's algorithm** to compute it
- From each station to all other stations

> **Tip: We can save the results in a new set $\mathcal{C}_{o,d}$**
> This will help us to model the movement inside the metro.

## Shortest Paths (SP)

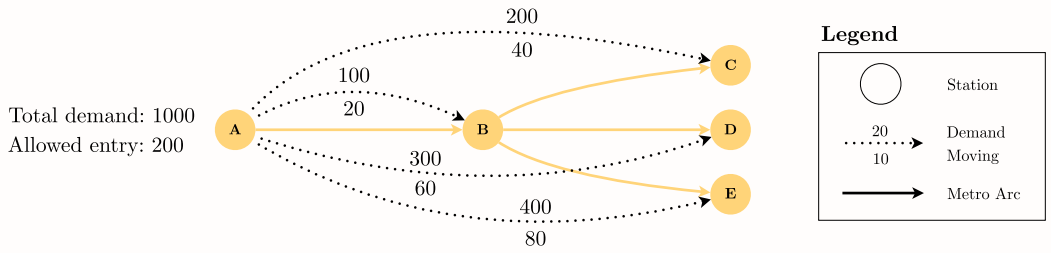
- $\mathcal{C}_{o,d}$ - Set of arcs $e \in \mathcal{E}$ on the SP from $o$ to $d$ with $o, d \in \mathcal{O}$

> **Note**
> The arc travel times $\tau_e$ are only the **input** to Dijkstra's algorithm. From the resulting paths we compute the travel time parameter the model actually uses:

- $\tau_{o,e}$ - Travel time (min) on the SP from station $o \in \mathcal{O}$ to arc $e \in \mathcal{E}$

## People Spreading

**Question: How do people spread?**




## Ratio of Origin-Destination

- We **cannot control** the destination of the passengers
- Thus we assume that people spread **based on destination**

$$\frac{q_{o,d,p}}{\sum_{f \in \mathcal{O}} q_{o,f,p}} \quad \forall o, d \in \mathcal{O}, p \in \mathcal{P}$$

> **Note**
> Based on the ratio of the different destinations $d$ to the total queue for each station $o \in \mathcal{O}$ in each period $p \in \mathcal{P}$. Note that the summation needs its **own index** $f$ over all destinations, as $d$ is already taken by the ratio's destination.

# Variables & Objective

## Decision Variable?

> ! **Our goal is to:**
> Regulate the **inflow** at **each individual station** to **minimize the queues** outside the metro stations based on the allowed inflow while **adhering to the capacity constraints.**

> **Tip**
> There is **one queue for all passenger flow directions**, as managing multiple queues would be too complex for the planners!

## Decision Variable

> i **We need the following sets:**
> - All the metro stations, $o \in \mathcal{O}$
> - All periods under observation, $p \in \mathcal{P}$

**Question: What could be our decision variable?**

- $X_{o,p}$ - Allowed inflow (per minute) at station $o$ in period $p$

## Objective Function?

> ! **Our main objective is to:**
> Regulate the **inflow** at **each individual station** to **minimize the queues** outside the metro stations based on the allowed inflow while **adhering to the capacity constraints.**

**Question: How again are queues minimized?**

- By the allowed inflow $X_{o,p}$ subtracted from the queue

## Objective Function

> i **We need the following parameters and variables:**
> - $q_{o,d,p}$ - People queued to travel from station $o$ to $d$ with $o, d \in \mathcal{O}$ in period $p$
> - $X_{o,p}$ - Allowed inflow (per minute) at station $o$ in period $p$

**Question: What could be our objective function?**

$$\text{Minimize} \quad \sum_{o \in \mathcal{O}} \sum_{p \in \mathcal{P}} \left( \sum_{d \in \mathcal{O}} q_{o,d,p} - m \times X_{o,p} \right)$$

## Objective Function Revisited

**Question: What does the solver really maximize here?**

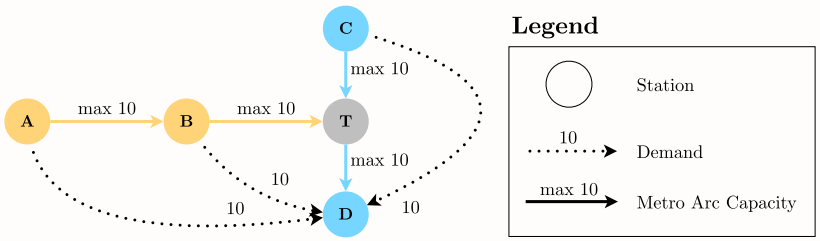
- The queued passengers $q_{o,d,p}$ are **parameters**, not variables
- Their sum is a **constant** — the solver cannot change it

> **Note: Minimizing queues = maximizing admissions**
> Minimizing the objective is exactly the same as **maximizing the total admitted passengers** $\sum_{o \in \mathcal{O}} \sum_{p \in \mathcal{P}} m \times X_{o,p}$. This is also why the model gets away without explicit queue dynamics.

# Constraints

## Necessary Constraints

**Question: What constraints do we need?**




- The capacity of each arc is **not exceeded**
- Do not dispatch more people than are queued
- Do not dispatch less than the minimum allowed inflow

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Things are going to

get a little

complicated now!

## Central Question

**Question: When do people flowing into the metro system change the arcs?**

- People enter metro station at $o$ with destination $d$
- They will lead to a usage of an arc $e$ on their SP
- This usage depends on their path and the travel times

> **Tip: You'll likely know what we need now!**
> We can add a new set $\mathcal{R}_{e,t}$ to help us with this.

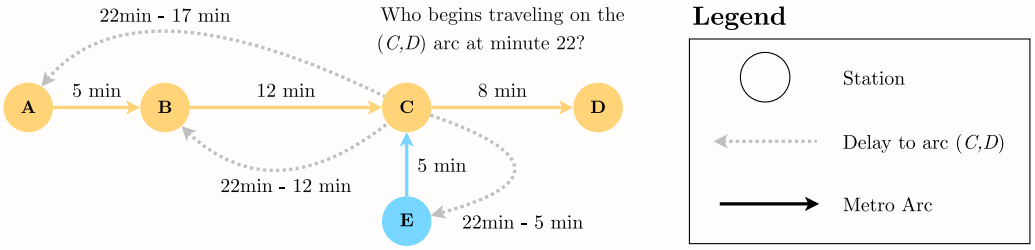
## Set of Time-Delays

$$\mathcal{R}_{e,t} = \{(o,d,p) \mid o,d \in \mathcal{O}, p \in \mathcal{P}, q_{o,d,p} > 0, \\ e \in \mathcal{C}_{o,d}, t - \tau_{o,e} \in I_p\} \quad \forall e \in \mathcal{E}, t \in \mathcal{T}$$

**Question: Who can explain this set?**

The set $\mathcal{R}_{e,t}$ contains **all combinations** $(o,d,p)$ which trigger a **capacity utilization** of arc $e$ in minute $t$.

## Small Example




> **Tip**
> The set contains all o-d pairs and periods that result in passengers entering arc $(C,D)$ at minute 22. With $m = 2$, it is $\{(A,D,3),(B,D,5),(C,D,11),(E,D,9)\}$.
>
> Check it yourself, e.g. from $A$: $\tau_{A,(C,D)} = 5 + 12 = 17$ and $22 - 17 = 5 \in I_3 = \{5,6\}$.

Essentially, we can use this set to compute the utilization of an arc at each minute based on the inflow level at the different stations and periods.

## Ensure Capacity Utilization?

> **! The goal of this constraint is to:**
> Ensure that the capacity of each arc is not exceeded at any minute.

**We need the following variable, parameter and set:**

- $q_{o,d,p}$ - people queued to travel from station $o \in \mathcal{O}$ to station $d \in \mathcal{O}$ in $p$
- $c_e$ - people max. allowed to enter arc $e$ per minute with $e \in \mathcal{E}$
- $\mathcal{R}_{e,t}$ - mapping of station entries to arc $e$ in minute $t$ with $e \in \mathcal{E}$ and $t \in \mathcal{T}$
- $\alpha$ - safety factor applied to all capacities $(0 < \alpha < 1)$
- $X_{o,p}$ - the allowed inflow per minute at metro station $o$ in the period $p$

## Ensure Capacity Utilization

**Question: What could be the constraint?**

$$\sum_{(o,d,p) \in \mathcal{R}_{e,t}} X_{o,p} \times \frac{q_{o,d,p}}{\sum_{f \in \mathcal{O}} q_{o,f,p}} \leq \alpha \times c_e \quad \forall e \in \mathcal{E}, t \in \mathcal{T}$$

**Note**

Here, we combine the inflow at each station based on the o-d proportion and let the people spread through the network, checking that no arc is over-utilized at any minute.

## Bound the Inflow Rate?

**The goal of this constraint is to:**

Ensure that the inflow rate at each station stays within operational bounds.

**We need the following variables and parameters:**

- $X_{o,p}$ - Allowed inflow (per minute) at station $o$ in period $p$
- $c_o^{min}$ - Min. station entry rate $o$ per minute
- $c_o^{max}$ - Max. station entry rate $o$ per minute
- $\alpha$ - Safety factor applied to all capacities $(0 < \alpha < 1)$

## Bound the Inflow Rate

**Question: What could be the constraint?**

$$c_o^{min} \leq X_{o,p} \leq \alpha \times c_o^{max} \quad \forall o \in \mathcal{O}, p \in \mathcal{P}$$

**Note**

This constraint bounds the inflow rate at each station between the **minimum and maximum** allowed entry rates, ensuring compliance with **operational capacity limits**.

## Dispatch Only Available People?

**The goal of this constraint is to:**

Ensure that we do not dispatch more people than are queued.

> **We need the following variables and parameters:**
> - $X_{o,p}$ - Allowed inflow (per minute) at station $o$ in period $p$
> - $q_{o,d,p}$ - People queued to travel from station $o$ to $d$ with $o, d \in \mathcal{O}$ in $p$
> - $m$ - period length in minutes

### Dispatch Only Available People

**Question: What could be the constraint?**

$$\sum_{d \in \mathcal{O}} q_{o,d,p} - m \times X_{o,p} \geq 0 \quad \forall o \in \mathcal{O}, p \in \mathcal{P}$$

> **Note**
> This constraint ensures **non-negative remaining queues**, preventing the solver from dispatching more passengers than available in the queue.

### A Hidden Pitfall

**Question: What if a queue is smaller than $m \times c_o^{min}$?**

- The lower bound forces $m \times X_{o,p} \geq m \times c_o^{min}$
- But we may only dispatch up to $\sum_{d \in \mathcal{O}} q_{o,d,p}$ people
- Example: $c_o^{min} = 2$, $m = 15$, but only 20 people queued
- A quiet station late at night makes the model **infeasible**!

**Question: How could we fix this?**

- Set $c_o^{min} = 0$, cap the bound at the queue, or soften the constraint

### Metro Inflow Model

subject to:

## Model Characteristics

### Characteristics

**Questions: On model characteristics**

- Is the model formulation linear/ non-linear?
- What kind of variable domains do we have?

> **Note: The answers**
> It is a **linear program**, as objective and constraints are linear. The variable $X_{o,p} \geq 0$ is **continuous** — fractional inflow rates per minute are perfectly acceptable here.

### Model Assumptions

**Questions: On model assumptions**

- What assumptions have we made?
- What are likely issues that can arise if applied?
- Have we thought in detail about queues?
- Are shortest paths a feasible assumption?

# In Practice

## Metro Inflow Optimization

**Question:** Can this be applied?

### Metro Inflow Problem

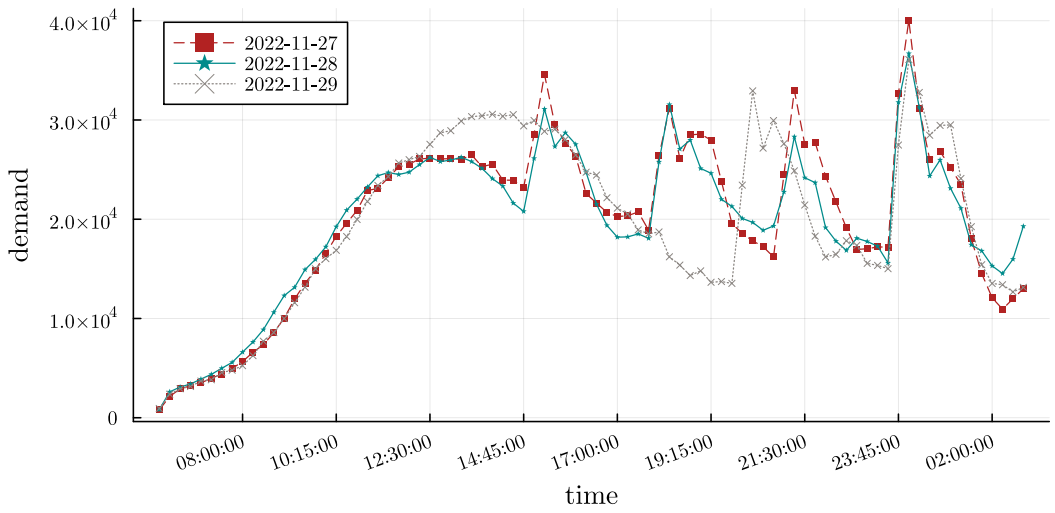
- Solved very fast **within seconds** for realistic problem sizes
- **But** we cannot plan or control the metro inflow
- Queues are **too simplified** with passengers disappearing

**Question: Any ideas, how the current model could be improved or how it could be embedded into a heuristic?**

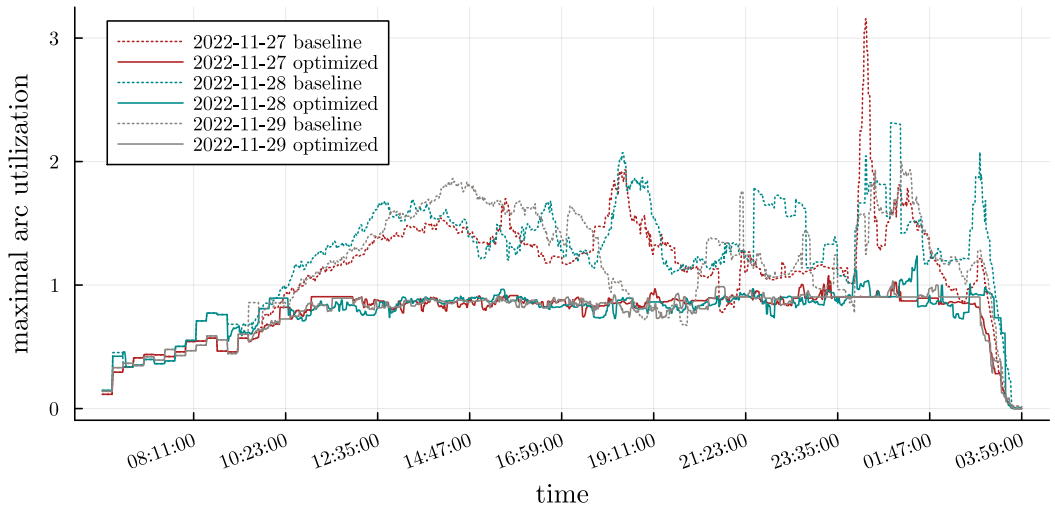
### Heuristic: Step-Wise

- Solve the model for the time-horizon of a **few periods**
- Fix the inflow in the **current first period**
- Decrease capacity in the network **based on the inflow**
- Transfer **remaining queues** into the subsequent period
- **Solve the model again**
- **Repeat**, until the inflow is computed for all periods

## Transport Demand




## Utilization Analysis




## Key Results

- Model reduces capacity violations from **25,941 to just 4**
- Maximum arc utilization drops from **316% to 109%**
- **81% of passengers** accommodated safely
- Computation time: **~1.4 seconds** per planning period

> **Note: Safety vs. Throughput Tradeoff**
> The remaining 19% of passengers experience external queuing - this is the price for preventing dangerous overcrowding inside the system.

## Didn't We Bound the Utilization?

**Question: Our constraint enforces $\alpha \times c_e$ with $\alpha < 1$. How can utilization still reach 109%?**

- The constraint binds the model for the **forecast** demand
- The 109% occurs in the **rolling-horizon simulation** with realized demand
- Realized flows can deviate from the forecast — but far less than **316%**

## An Ordinary Day in Shanghai

Question: Does inflow control also work without a World Cup?

## Scalability: Shanghai Metro

- Validated on **302 stations** (~8× more than Doha's 37)
- **99% reduction** in capacity violations
- Computation: **< 2 minutes** per period
- Works for routine **peak-hour demand**

> **Tip: Generalizable Approach**
> The method scales from event-driven demand (Qatar World Cup) to routine metropolitan operations (Shanghai rush hour).

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## Implementation

- **Assumption of known destinations** is strong
- Movements **seemed to follow our forecasts**
- We did achieve our goal of **metro inflow control**
- Simulation was used to **estimate the inflows**

> **Note**
> Few dangerous situations, especially at the FIFA Fan Fest, were **handled well** by the authorities.

## Comparison with Others

| Feature | Other Methods | Our Approach |
|---|---|---|
| FIFO Enforcement | Rarely | **Yes** |
| Single Queue | Often separate | **Yes** |
| Real-Time Capable | Some | **Yes (seconds to minutes)** |
| Network Scale | Up to 327 stations | Comparable (302 stations) |

> **Note: State-of-the-Art**
> First network-level approach combining FIFO enforcement, single-queue modeling, and real-time computational performance.

### Wrap Up

- Model can help to achieve a **good balance**
- Can be adapted easily to **any metro system worldwide**
- Especially interesting for **larger Asian cities**

> **And that's it for today's lecture!**
> We now have covered a metro inflow control problem based on a real-world application and are ready to start solving some new tasks in the upcoming tutorial.

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Questions?

## Literature

For more interesting literature to learn more about Julia, take a look at the literature list of this course.